Tissue organisation, communication, and interpretation

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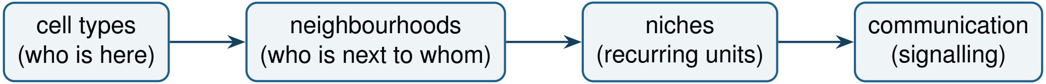
Today, and why it stands alone

- We run on a separate, smaller machine (6 GB) and a small, pre-annotated single-cell spatial dataset (seqFISH mouse embryo). Nothing today depends on Days 1–2.
- Focus shifts from *making* the data to *reading* it: organisation, communication, and how to interpret each spatial statistic in practice.
- For every statistic we ask three things: what it measures, how to read the output, and where it can mislead.

From cell types to tissue organisation

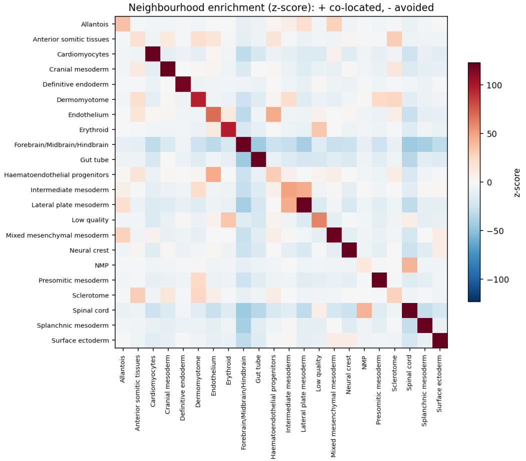


Each step is built from the spatial graph. Proportions alone miss the arrangement, which is where tissue function lives.

Everything depends on the spatial graph

- Enrichment, co-occurrence, niches, autocorrelation, and communication are all computed *on a graph* of which cells count as neighbours.
- The graph is a choice: kNN (fixed count), Delaunay (tissue geometry), or a fixed radius. We use Delaunay today.
- **Consequence:** every result is conditional on that choice. Change the graph and the numbers change, so state it and, when it matters, check robustness to it.

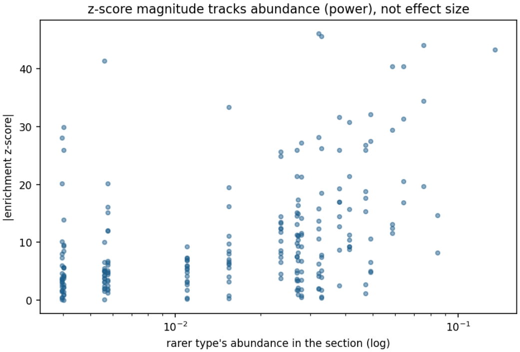
Neighbourhood enrichment: how to read it



Permute cell-type labels on the graph; compare observed adjacency to the null, reported as a **z-score**.

- **Diagonal**: how strongly a type sits next to itself (spatial aggregation). Almost always strongly positive here.
- **Off-diagonal** > 0 (red): the two types co-locate; < 0 (blue): they avoid each other.
- Read pairs, e.g. Endothelium $\leftrightarrow$ Haematoendothelial progenitors (positive), brain $\leftrightarrow$ gut (negative).

Neighbourhood enrichment: limits of the z-score

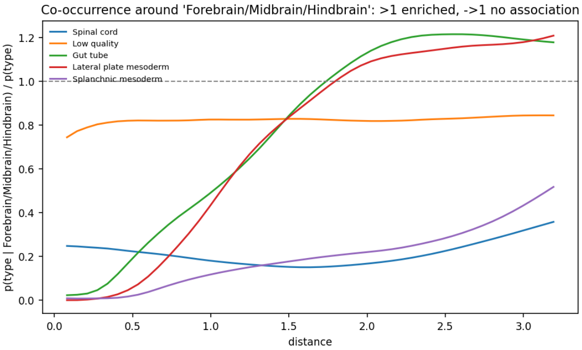


The z-score is a *significance* measure, not an effect size.

- It grows with the number of cells, so abundant pairs reach large $|z|$ from small effects (above).
- It is relative to this section's **composition**: a difference between samples can be a composition difference, not a real change in arrangement.
- It depends on the graph.

So rank pairs by $|z|$ *within* a section, but do not compare raw z across sections or read it as strength.

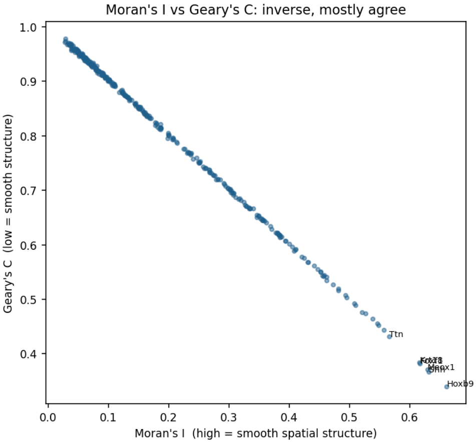
Co-occurrence: how to read it



The curve is $p(\text{type} \mid \text{focus}, d)/p(\text{type})$ against distance d:

- **above 1**: enriched at that distance; $\to 1$: no association; **below 1**: depleted.
- **Shape matters**: high at short d = direct contact; rising only at large d = separate compartments (brain vs gut here).
- Complements enrichment: enrichment is one number, co-occurrence is the distance profile behind it.

Is a gene spatially structured? Moran’s I and Geary’s C



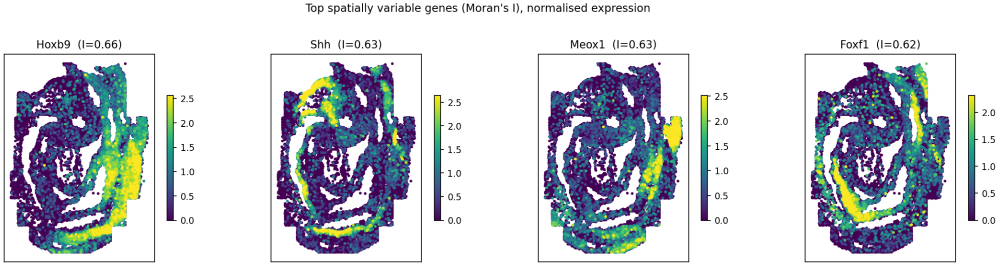
Both rank genes by how spatially structured their expression is.

$$I = \frac{N}{W} \frac{\sum_{ij} w_{ij}(x_i - \bar{x})(x_j - \bar{x})}{\sum_i (x_i - \bar{x})^2}$$

- **Moran’s I**: ≈ 1 smooth structure, ≈ 0 random, < 0 checkerboard. Global, gradient-sensitive.
- **Geary’s C**: ≈ 0 structure, ≈ 1 random. More local.
- They are inverse and mostly agree (left); use I for a first pass.

Moran 1950; Geary 1954.

Spatially variable genes in space



A high Moran's I says a gene is spatially structured, not *what* structure. Always plot the top hits: here they mark discrete anatomy. A gene high in one small region can still score low globally, and a high I does not imply cell-type specificity, so pair the ranking with the maps.

Highly variable vs spatially variable genes

Highly variable (HVG)

- “Varies a lot across cells”, ignoring location.
- One scalar per gene: variance, or dispersion (variance/mean), or the scanpy HVG score.
- Mean-corrected and directly comparable, so ranking is straightforward.

Spatially variable (SVG)

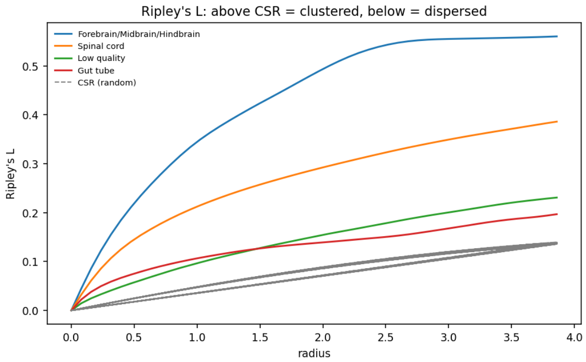
- “Varies in an organised way *in space*”.
- Scored by Moran’s I (or SpatialDE, SPARK-X, . . .), which needs the spatial graph.
- The score is not mean-corrected, so it is not directly comparable across genes, see next.

A gene can be highly variable but not spatial (noisy everywhere), or spatial but not highly variable (a smooth, low-amplitude gradient).

Comparing genes by Moran's I

- Moran's I measures *spatial smoothness* on a fixed graph, not how strong or how specific a gene is. A high I can come with a small fold change.
- It is influenced by **mean expression and detection rate**: sparse, lowly expressed genes look noisier and score lower, so a raw I ranking favours abundant, broadly expressed genes.
- So do not read a small I difference between two very different genes as "more spatial". Instead:
  - use the **permutation p-value** (FDR-corrected) to ask *whether* a gene has spatial structure;
  - compare strength only **within similar expression levels**, or on variance-stabilised values;
  - use I to shortlist, then **plot and read in space**.

Ripley's K / L: clustering across scales



For one cell type, count how many same-type cells fall within radius r, vs a random (CSR) expectation.

- **above CSR**: clustered at that scale; **below**: dispersed/regular.
- The L transform stabilises the variance; the gap from CSR is the effect.
- The *scale* where curves separate tells you how big the structures are (brain clusters over a wide range here).

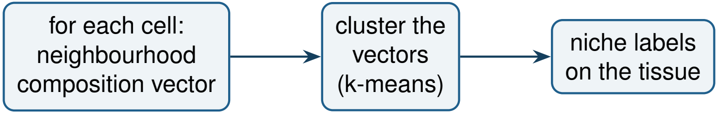
Niches and tissue domains: the method landscape

- **Composition clustering** (what we do): summarise neighbourhood cell-type composition, then cluster. Transparent, fast, CPU-friendly.
- **BANKSY**: augments each cell with its neighbourhood; types cells and finds domains together, scalable and batch-aware.
- **CellCharter**: clusters neighbourhood-aware embeddings into niches across samples.
- **SpaGCN, STAGATE**: graph neural networks for spatial domains.

Learned methods can capture subtler domains; they add assumptions and compute.

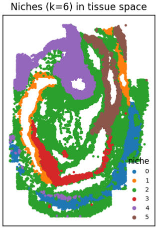
Singhal et al., *Nat Genet* 2024 (BANKSY); Varrone et al., *Nat Genet* 2024 (CellCharter); Hu et al., *Nat Methods* 2021 (SpaGCN); Dong & Zhang, *Nat Commun* 2022 (STAGATE).

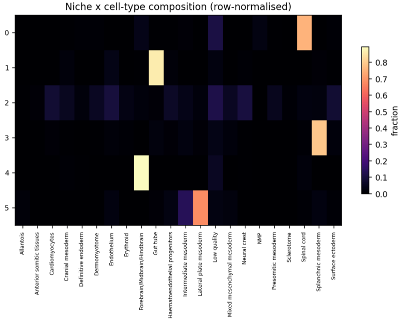
Building niches you can interpret



The output is named, mappable units you can compare across samples, with no black box between the data and the label.

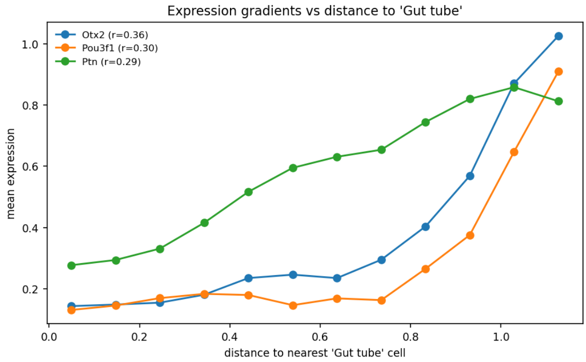
Reading niches: the composition is the label





- Each niche *is* its row in the composition heatmap: niche 1 = gut tube, niche 4 = brain, niche 0 = spinal cord.
- A row dominated by one type is a homogeneous compartment; a mixed row is an interface or a generic background niche.
- **Choosing** k: scan a few values, keep niches that are spatially coherent and stable, not salt-and-pepper. There is no true k.

Distance-to-structure gradients



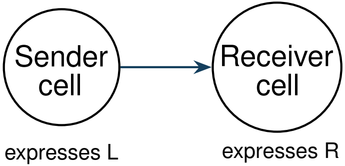
Pick a structure (here the gut tube), give every cell its distance to the nearest such cell, and read expression against distance.

- Turns “where” into a continuous covariate, exposing gradients that clusters and niches miss.
- Genes rising away from the gut (Otx2, Pou3f1) trace the neural compartment.
- Generalises to any landmark: a vessel, a tumour boundary, a wound edge.

Cell–cell communication

Score ligand–receptor pairs between cell-type groups: is the ligand expressed by one type and the receptor by a co-located type?

In spatial data we can add the constraint that the partners are actually neighbours, which dissociated data cannot.



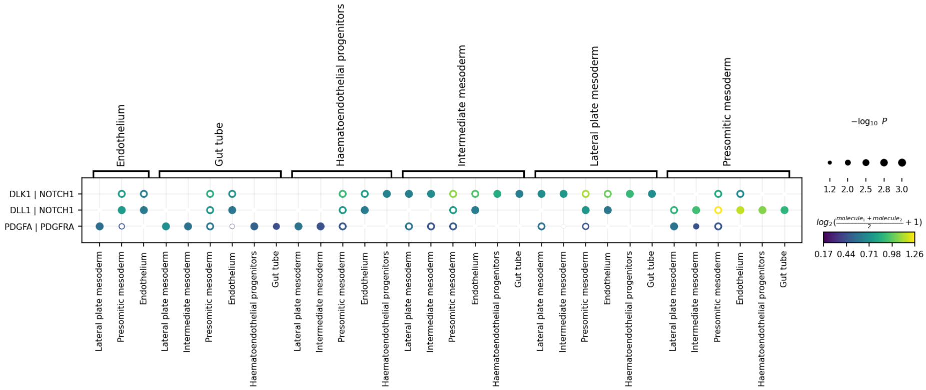
Communication: methods and caveats

- **Non-spatial:** CellPhoneDB, CellChat, NicheNet (adds downstream targets); LIANA unifies many under one interface.
- **Spatial-aware:** squidpy `ligrec`, stLearn, COMMOT (optimal transport), NCEM (niche effects on expression).

Caveats: expression is a proxy for activity; a small panel covers few pairs; p-values depend on the permutation scheme; segmentation errors move ligands to the wrong cell. Read results as hypotheses.

Efremova et al., *Nat Protoc* 2020 (CellPhoneDB); Jin et al., *Nat Commun* 2021 (CellChat); Browaeys et al., *Nat Methods* 2020 (NicheNet); Dimitrov et al., *Nat Commun* 2022 (LIANA); Cang & Nie, *Nat Methods* 2023 (COMMOT); Fischer et al. 2023 (NCEM).

Reading a ligand–receptor result



Columns are sender → receiver cell-type pairs (grouped by sender), rows are ligand | receptor pairs. **Dot colour** = mean expression of the pair across the two types; **dot size** = $-\log_{10} p$ (permutation); **filled** = significant. A big, bright, filled dot is the strongest hypothesis (e.g. DLL1 | NOTCH1 into presomitic mesoderm). It is a candidate, not proof of signalling.

Significant is not the same as important

- **Significant?** (p-value) vs **how big?** (effect size) vs **does it matter?** (biology). A tiny effect can be very significant with enough cells, as the enrichment z-scores showed.
- **Multiple testing**: hundreds of genes and many cell-type pairs mean many tests. Correct (Benjamini–Hochberg / FDR) and report it.
- **Permutation nulls** answer a specific question (labels shuffled on a fixed graph): the p-value inherits that null's assumptions.
- n **is cells, not samples**: significance from one section does not generalise to the condition. For that you need replicates.

Confounders to keep in mind

- **Composition**: differing cell-type proportions masquerade as different arrangement. Compare like with like.
- **The graph**: kNN vs Delaunay vs radius changes every downstream number.
- **Density and edges**: cell density gradients and the section boundary bias neighbour-based statistics.
- **Segmentation and panel** (from Days 1–2): misassigned transcripts corrupt co-expression and communication; low cytokine detection makes absence weak evidence.

Treat every result as a hypothesis conditioned on these choices.

Comparing conditions

- With a disease cohort, the question becomes which niches or compositions change between groups.
- Composition-aware tests (e.g. Milo for differential abundance, scCODA for composition) avoid the pitfalls of comparing raw proportions.
- Spatial features (niche burden, distances, enrichment scores) can be summarised per sample and tested like any covariate, with the sample as the unit of replication.

Dann et al., *Nat Biotechnol* 2022 (Milo); Büttner et al., *Nat Commun* 2021 (scCODA).

The frontier

- Integrating large H&E images with the molecular layer, and pathology image foundation models for region identity (e.g. UNI, Virchow, H-optimus), with attention to their failure modes.
- Multimodal imaging plus Xenium to give better identity to regions of interest.
- Agentic and LLM-assisted analysis workflows.

These are open and moving quickly; today's tools are a foundation, not the ceiling.

Further reading

- Varrone et al. CellCharter. *Nat Genet* 2024; Singhal et al. BANKSY. *Nat Genet* 2024.
- Cang & Nie. COMMOT. *Nat Methods* 2023; Dimitrov et al. LIANA. *Nat Commun* 2022.
- Dann et al. Milo. *Nat Biotechnol* 2022.
- Squidpy tutorials (neighbourhood enrichment, co-occurrence, Ripley, ligrec).

Hands-on: day_3/03_tutorial.ipynb

Dataset: a small pre-annotated seqFISH section. You will:

- run neighbourhood enrichment and co-occurrence, and read them;
- identify niches from neighbourhood composition;
- measure a distance-to-structure gradient and rank spatially variable genes;
- score ligand-receptor interactions.